FLOATING DJ VISUALIZER

V3.0.0 · IMPLEMENTATION REPORT

MODULAR AUDIO EQUALIZER ENGINE & FLOATING PROFILE SCROLL FIX

Baseline: V2.4.32 | Target device: Redmi 10C / Android 11 (API 30)
minSdk 26 · targetSdk 36 · CodeAssist CAPROJ

BUILD SCOPE: EQ Activity · Capability Probe · Effect Controller · Presets · Route Monitor · Engine Service

Prepared 13 November 2026

Engineering status: source integration completed; Android build/device proof pending

01 · Executive summary

V3.0.0 keeps the V2.4.32 Visualizer renderer and FloatingVisualizerService source as the baseline. The update adds an isolated EQ entry point and an experimental Android AudioEffect path, while correcting nested vertical scrolling inside Floating Profile. The implementation deliberately reports actual engine capability instead of pretending that every phone provides a 15-band global equalizer.

Important engineering boundary
A successful AudioEffect constructor, a moving slider, or an ON switch is not proof that all device audio is being processed. Global output-mix insert effects using audio session 0 are deprecated by Android documentation. Device-wide post-processing requires vendor/OEM support and is not a general public permission for ordinary APKs.

Delivered in the source package

Area	V3.0.0 change	Status
Main UI	EQ · Equalizer Audio Engine card added beneath Floating Overlay controls. Opens a separate EqualizerActivity.	Integrated
Floating Profile scroll	Inner profile ScrollView disallows parent interception from touch-down through gesture completion. Outer page remains independently scrollable outside that canvas.	Integrated
EQ engine	Session-0 Equalizer/BassBoost/Virtualizer experimental controller; capability inventory, hasControl and parameter read-back.	Experimental
Preset manager	Save, Save As, Rename, Delete, Flat reset, JSON import/export with schema and range validation.	Integrated
Lifecycle	Foreground EqEngineService holds the selected effect while enabled; notification explains experimental scope.	Integrated
Reverb / limiter	Not falsely exposed as global active effects. Reverb requires an appropriate auxiliary player/send path; limiter requires supported processing engine.	Deferred

02 · Project structure

The project keeps the existing Visualizer package and adds the EQ subsystem under its own namespace. MainActivity only launches the EQ screen; audio-effect ownership is not placed in the renderer or FFT loop.

File / component	Responsibility
EqualizerActivity.kt	EQ interface, actual native band controls, bass and virtualizer controls, preset actions and SAF import/export.
eq/core/EffectCapabilityProbe.kt	Read-only AudioEffect descriptor inventory: effect name, implementor, type UUID, implementation UUID and connect mode.
eq/core/EqEngineController.kt	Single owner for Equalizer/BassBoost/Virtualizer handles; session-0 experiment, control checks, parameter read-back and release.
eq/service/EqEngineService.kt	Foreground lifecycle owner for the user-enabled experiment; restarts only when the stored enabled state requests it.
eq/model/EqProfile.kt	Portable preset model with center frequencies, gain values, BassBoost and Virtualizer state.
eq/data/EqPresetRepository.kt	Local preset storage and schemaVersion 1 JSON import/export.
eq/route/EqAudioRouteMonitor.kt	Reports connected output categories: built-in speaker, wired headset, Bluetooth and USB audio.

Audio control flow

MainActivity → EQ card → **EqualizerActivity** → capability probe → one EQ controller → Android AudioEffect framework → vendor effect implementation / audio route. The EQ controller is independent of the Visualizer FFT and floating renderer.

The EQ controller is not recreated on every slider movement. Slider updates mutate the existing handle and display the value read back from the effect. If control is unavailable or lost, the UI must not label the effect as successfully active.

03 · Audio engine and feature truthfulness

Android exposes Equalizer, BassBoost, Virtualizer, PresetReverb, EnvironmentalReverb and DynamicsProcessing classes. The existence of a class does not mean that a device exposes a matching implementation, that the app owns its control, or that the effect applies to every app and output route.

Feature	V3.0.0 behavior	Boundary
Equalizer	Reads native band count, center frequency and level range. Displays only the bands returned by the actual effect.	15 target points are not fabricated when device reports fewer bands. Session 0 is deprecated.
Bass Boost	Uses native BassBoost only if created and controlled; reads rounded strength.	Not equivalent to 15-band EQ. Strength may be quantized by vendor implementation.
Virtualizer	Uses native Virtualizer only if available and controllable.	Route-dependent; API deprecated from API 35. Spatializer is not a generic EQ replacement.
Reverb	Displayed as a deferred capability, not as an active global control.	Preset/Environmental Reverb are auxiliary effects; player must provide the matching send path.
Limiter / dynamics	Not claimed active in this build.	DynamicsProcessing requires API 28+ and an actual supported session/effect engine.

Why session 0 is experimental

Android API reference says insert effects (Equalizer, BassBoost and Virtualizer) attached to the global output mix using session 0 are deprecated. The app therefore labels this mode LEGACY OUTPUT MIX / EXPERIMENTAL and makes no universal system-wide claim. A successful control check still needs route-by-route audio proof.

Direct kernel or DSP writes are not performed. The normal Android audio path is application → framework / AudioFlinger → Effects HAL / Audio HAL → vendor driver, codec or DSP. Access to public AudioEffect APIs is not equivalent to privileged kernel control.

04 · iJoysoft reference review

The supplied still image and screen recording were treated as UI and interaction references. Public Google Play descriptions identify iJoysoft equalizer apps with a five-band equalizer, Bass Boost, Virtualizer, preset management and, for one listing, Preset Reverb. The screenshot supplied by the project visibly shows six fader columns in the captured viewport; that image alone does not establish the full band count or internal implementation.

The app package source, native libraries, AudioEffect session IDs, vendor-specific calls and audio route behavior are not exposed by the public listing or the recording. Therefore, no claim is made that iJoysoft writes directly to the kernel. The only reliable way to characterize its effect path on the test phone would be a controlled runtime experiment and before/after output measurement.

Reference comparison

Observed / published	What we can infer	What remains unknown
Five-band EQ in Google Play description.	The published app description advertises five bands for the referenced listing.	Whether another iJoysoft variant/version uses a different band count.
Bass Boost, Virtualizer, presets and Reverb are listed in product descriptions.	These are the feature categories to use as product references.	Which Android effect class, session scope or vendor path is used on this specific Redmi 10C.
Screen recording shows active sliders and sound controls.	The UI interaction and visual style can be used as design reference.	Whether every slider changes the whole device mix or only selected streams.

Design decision

The EQ panel uses a dark blue surface, native vertical faders, frequency labels, value readouts, preset selector, close control, Bass Boost and Virtualizer sections. Unlike a static mockup, the number of rendered faders is derived from the native effect at runtime.

05 · Floating Profile scroll fix

The V2.4.32 layout uses an outer MainActivity ScrollView and a fixed-height inner ScrollView named floatingProfilesScroll. A vertical drag that begins inside the inner list can be intercepted by the parent page unless the child requests that ancestors not intercept the gesture.

V3.0.0 installs a touch listener on floatingProfilesScroll. It requests parent interception lock on ACTION_DOWN / ACTION_MOVE and releases the lock on ACTION_UP / ACTION_CANCEL. Returning false preserves the ScrollView's own touch handling. This is scoped only to the Floating Profile canvas; it does not globally disable the main page scroll.

Expected behavior
When the user drags inside the Floating Profile list, only that list scrolls. When the user starts a gesture elsewhere on the MainActivity page, the outer page continues to scroll normally. Reordering gestures on profile drag handles remain handled by the existing profile list code.

06 · Test plan and acceptance gates

Test	Procedure	Pass condition
Build / install	Import CAPROJ in CodeAssist, run clean build, install on Android 11.	No compile/resource/manifest errors; app launches without crash.
Visualizer regression	Test Single and Horizontal modes, Display preview, floating overlay, profiles, themes and notifications.	V2.4.32 behavior remains intact; no EQ action restarts FFT/renderers.
Profile nested scroll	Create 4+ floating profiles. Drag inside the list; then drag elsewhere on the page.	Inner list scrolls independently; outer page scrolls only outside list.
Effect inventory	Open EQ and record descriptors, implementor, UUID, connect mode and constructor result.	Capability status accurately reflects device output.
Audio A/B	Use identical audio and route. Change one supported band substantially, compare bypass/on.	A measurable/audible change is attributable to the tested route.
Routes	Repeat speaker, wired headset and Bluetooth.	Each route is reported independently; no unsupported route is claimed.
Presets	Save, Save As, rename, delete, export, import and try malformed JSON.	Valid data round-trips; malformed or out-of-range data is rejected.
Control lifecycle	Enable, close UI, reopen, disable, revoke/change output route and test competing EQ apps.	No duplicate handles, crash, stuck active state or silent control-loss.

07 · Build status and limitations

Source integration: the V3.0.0 project package contains the EQ Activity, engine/controller, capability probe, service, preset repository/model, route monitor, MainActivity entry card and nested-scroll gesture fix. Version metadata is updated to 3.0.0 / versionCode 70.

Static checks performed: project archive structure inspected; AndroidManifest and layout XML parsed successfully; new Kotlin files passed syntax-level inspection for parser errors.

Not verified in this environment: full Android compilation, CodeAssist import, APK installation, runtime AudioEffect behavior, sound output changes and the Redmi 10C route matrix. No Android SDK platform jar or Gradle build tool was available in the build environment. Therefore this package must be treated as source-integrated but not device-certified.

Do not interpret this report as a successful device test.
Before release, import the CAPROJ into the same CodeAssist environment used for V2.4.32, run a clean debug build, install it on the Redmi 10C, then capture the EQ diagnostics and test speaker / wired / Bluetooth separately. If session-0 control is rejected or ineffective, the app must show unavailable rather than claiming global EQ.

Roadmap carried forward

V3.0.0-A: verify device capability and output-mix experiment. **V3.0.0-B:** stabilize EQ core and preset portability based on native bands. **V3.0.0-C:** Bass Boost, preamp/headroom and limiter only where a real engine supports them. **V3.0.1:** route-aware Spatializer / Virtualizer and Reverb. **Later:** evaluate supported OEM/privileged integration if truly global device processing is required.

Technical references

1. Android Developers — AudioEffect API: developer.android.com/reference/android/media/audiofx/AudioEffect
2. Android Developers — Equalizer API: developer.android.com/reference/android/media/audiofx/Equalizer
3. Android Developers — BassBoost API: developer.android.com/reference/android/media/audiofx/BassBoost
4. Android Developers — Virtualizer API: developer.android.com/reference/android/media/audiofx/Virtualizer
5. Android Developers — PresetReverb API: developer.android.com/reference/android/media/audiofx/PresetReverb
6. AOSP — Audio effects: source.android.com/docs/core/audio/audio-effects
7. iJoysoft — Equalizer & Bass Booster, Music (Google Play): Google Play listing
8. iJoysoft — Equalizer - Bass Booster & Volume (Google Play): Google Play listing

Prepared for the Floating DJ Visualizer project. Baseline source: V2.4.32 CAPROJ supplied by the project owner.